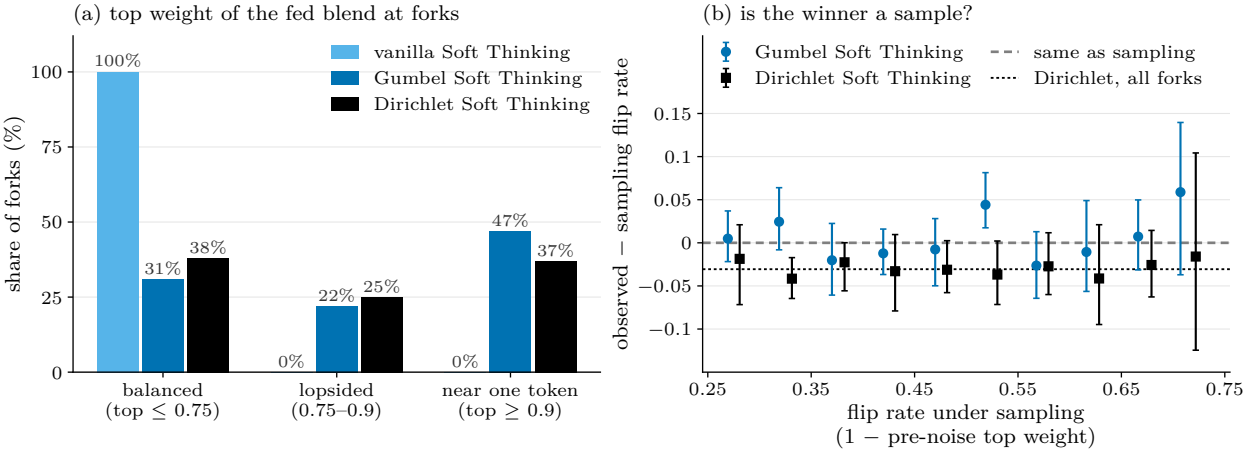

(a) top weight of the fed blend at forks
vanilla Soft Thinking
Gumbel Soft Thinking
Dirichlet Soft Thinking
share of forks (%)
100
75
50
25
0
100%
31%
38%
0%
22%
25%
0%
47%
37%
balanced
(top ≤ 0.75)
lopsided
(0.75–0.9)
near one token
(top ≥ 0.9)
(b) is the winner a sample?
Gumbel Soft Thinking
Dirichlet Soft Thinking
same as sampling
Dirichlet, all forks
observed − sampling flip rate
0.15
0.1
0.05
0
−0.05
−0.1
0.25
0.35
0.45
0.55
0.65
0.75
flip rate under sampling
(1 − pre-noise top weight)